POWER BI PROJECT REPORT

RAJALAKSHMI ENGINEERING COLLEGE

FINANCIAL PERFORMANCE ANALYSIS

Sales, profitability and market performance

PROJECT REPORT AND ANALYTICAL WORKINGS

NET SALES	TOTAL PROFIT	PROFIT MARGIN
$118.73M	$16.89M	14.2%

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FINANCIAL PERFORMANCE ANALYSIS

Power BI project documentation | Report design, data preparation and analytical workings

1. Project overview

The supplied PBIX is a sales and profitability report. It analyzes 700 transaction records across product, customer segment, country and date. This report documents the data, preparation logic, model structure, measures, report pages, validation checks and findings. The body covers the supplied sales and profitability project.

TRANSACTIONS	NET SALES	PROFIT	MARGIN
700	**$118.73M**	**$16.89M**	**14.2%**

Project objectives

- Prepare the sample for reliable analysis and preserve the source values.
- Model date and transaction data so month, year, product, country and segment filters behave consistently.
- Compare sales, cost of goods sold, profit and profit margin without aggregating incompatible categories.
- Explain the report pages and show how a reviewer can reproduce the analysis.

PBIX report map

Executive Overview summarizes KPI cards, profit trend, product/segment charts and slicers. **Profit & Loss Report** presents a year/month matrix and monthly sales/profit trend. **Profitability Analysis** compares profit by product and country and exposes transaction details. The PBIX definition also includes checks named Sales Check and Profit Check.

Scope note: the dataset is Microsoft demonstration data, not audited company accounts.

2. DATASET AND SCOPE

Source, time coverage and the fields used in the report

Dataset profile

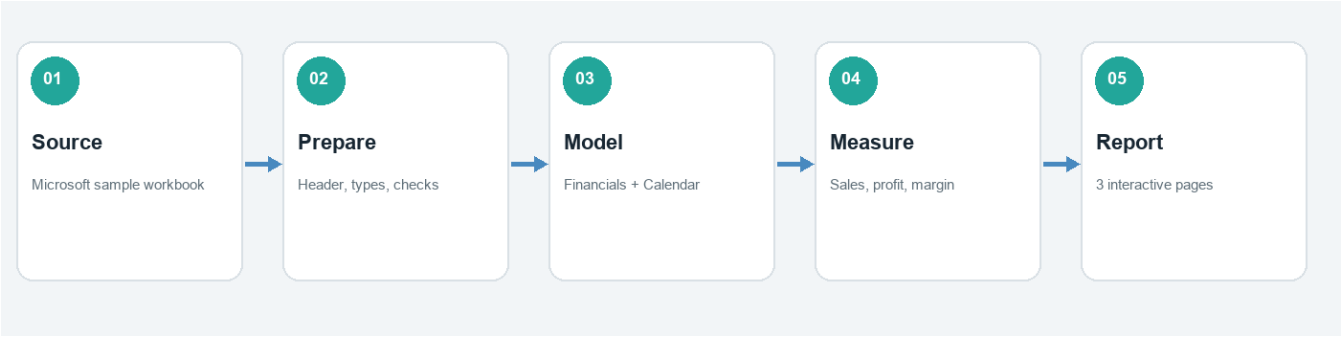
The compatible dataset contains **700 rows** and **16 fields**, covering 2013-09 through 2014-12 (16 months). It includes 5 countries, 6 products, 5 segments and 4 discount bands. There is one record per sample transaction row; no unique transaction ID is supplied.

Field	Type	Use
Segment	Text	Market/customer group
Country	Text	Market geography
Product	Text	Product category
Discount Band	Text	Discount category
Units Sold	Decimal	Quantity
Manufacturing Price	Currency	Unit manufacturing cost
Sale Price	Currency	List price per unit
Gross Sales	Currency	Before discounts
Discounts	Currency	Discount amount
Sales	Currency	Net sales; normalized header
COGS	Currency	Cost of goods sold
Profit	Currency	Sales less COGS
Date	Date	Transaction date
Month Number	Whole number	Calendar month 1-12
Month Name	Text	Calendar month label
Year	Whole number	Calendar year

Source: Microsoft Power BI Financial Sample workbook. The project dataset is distributed with this report as CSV and formatted XLSX. The workbook's source column " Sales" includes an initial blank; the prepared dataset renames it to "Sales" for consistent use in Power BI.

3. DATA PREPARATION AND QUALITY

Reproducible Power Query sequence and checks applied to the prepared extract



Preparation sequence

Step	Work performed / recommended
1. Source	Import the Financial Sample workbook / prepared CSV into a Financials query.
2. Headers	Promote the first row as headers and trim surrounding whitespace; rename “ Sales” to “Sales”.
3. Values	Keep text categories as text; parse Date as Date, Year and Month Number as whole numbers, and amount/quantity columns as numeric.
4. Missing values	Check blank cells and preserve true missing values as null; do not silently replace missing values with zero.
5. Model date	Create Calendar covering the minimum and maximum transaction dates; mark it as the date table.
6. Validate	Compare row counts and totals, verify sales/profit arithmetic, and check repeated candidate keys before removing any records.

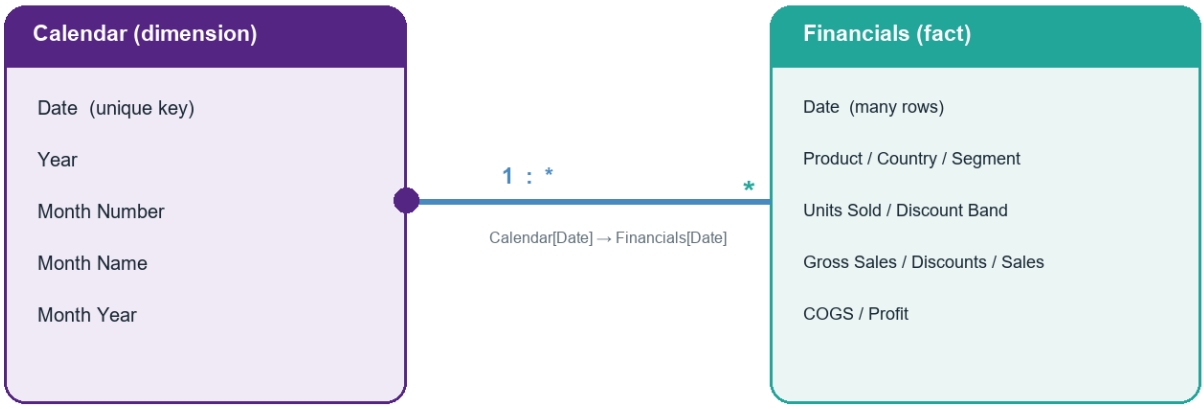
Data quality results

700 records; **0** blank cells across the 16 supplied columns. Arithmetic checks passed on all rows: Gross Sales - Discounts = Sales (0 exceptions) and Sales - COGS = Profit (0 exceptions), with one-cent tolerance. A candidate combination of Date, Segment, Country, Product and Discount Band repeats in 12 groups (12 excess rows); the source has no transaction ID, so repeated combinations are not automatically errors and should not be deleted without source confirmation.

4. DATA MODEL AND MEASURES

Separate calendar filtering from transaction facts; keep financial calculations reusable

Power BI relationship layout



Model design

Use **Calendar[Date]** as the one-side key and **Financials[Date]** as the many-side key in a single-direction relationship. Calendar provides Year, Month Number, Month Name and Month Year. Sort Month Name by Month Number and use a Year-Month sort key for chronological visuals. The Financials grain is transaction-level; slice or group by dimensions, then aggregate additive values.

DAX measure definitions for the cleaned CSV

Measure	Definition / purpose
Total Gross Sales	`Total Gross Sales = SUM(Financials[Gross Sales])`
Total Discounts	`Total Discounts = SUM(Financials[Discounts])`
Total Sales	`Total Sales = SUM(Financials[Sales])`
Total COGS	`Total COGS = SUM(Financials[COGS])`
Total Profit	`Total Profit = SUM(Financials[Profit])`
Profit Margin %	`Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)`
Total Units Sold	`Total Units Sold = SUM(Financials[Units Sold])`

Reconciliation logic: Sales Check = Total Gross Sales - Total Discounts - Total Sales; Profit Check = Total Sales - Total COGS - Total Profit. Expected result is zero. Format Profit Margin % as percentage; do not sum percentages or average dollar values across unrelated units.

5. EXECUTIVE OVERVIEW

How the report summarizes the full transaction history

The supplied PBIX places five KPI cards across the top: Total Profit, Total Sales, Profit Margin %, Total COGS and Total Units Sold. Below, it compares profit over time, profit by product and sales by segment. Country, year, segment and product slicers let the presenter narrow all visuals to a selected context.

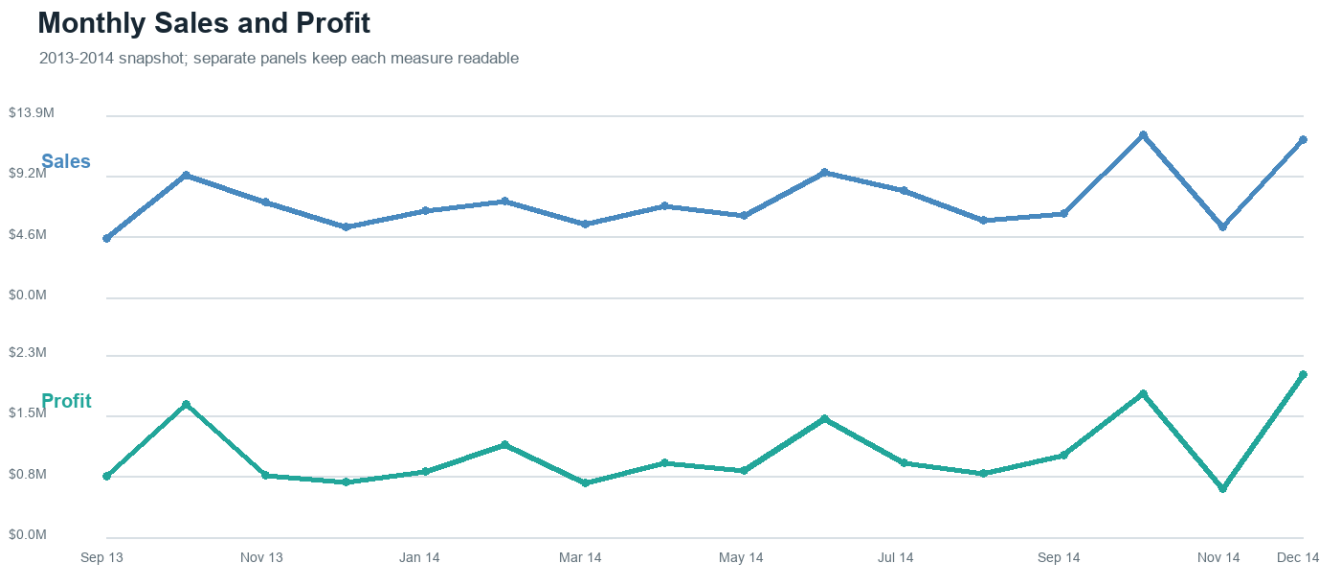


Figure 1. Monthly Sales and Profit preview. Separate panels are used here to keep both values readable; the PBIX visual can be viewed with its own filtering and formatting.

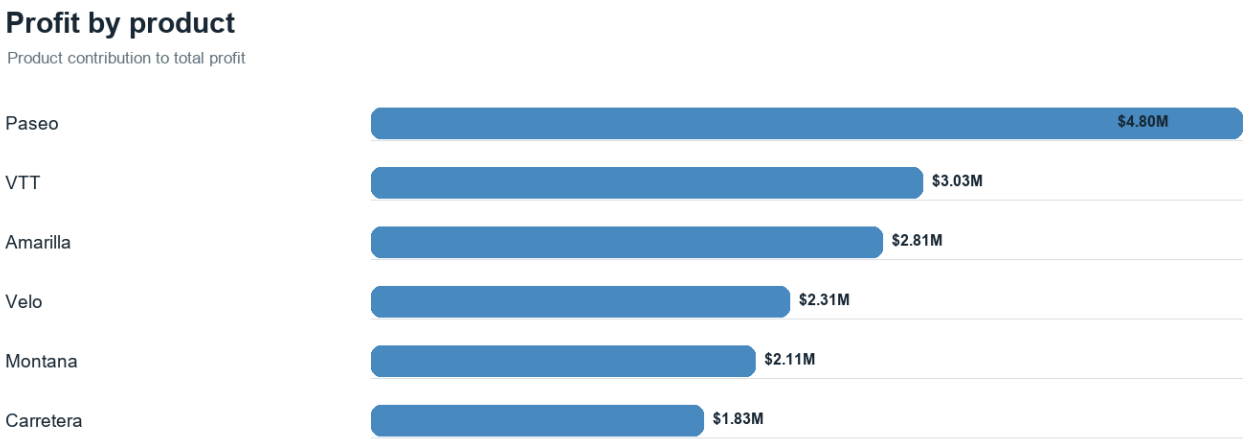


Figure 2. Product profit preview. Paseo is the largest profit contributor in the full sample.

6. PROFIT & LOSS AND PROFITABILITY

Detailed time view and comparisons across the business dimensions

Profit & Loss Report

The PBIX page uses a Year / Month matrix with Gross Sales, Discounts, Sales, COGS, Profit and Profit Margin, plus a monthly Sales and Profit line chart. It supports tracing annual totals down to months and comparing net sales with costs.

Amount bridge	Value
Gross Sales	$127,931,598.50
Discounts	$-9,205,248.24
Net Sales	$118,726,350.26
COGS	$-101,832,648.00
Profit	$16,893,702.26

Profitability Analysis

The PBIX page compares total profit by product and by country, then shows transaction detail with Country and Product slicers. The previews below show sample totals; a selected slicer changes the comparison to that subset.

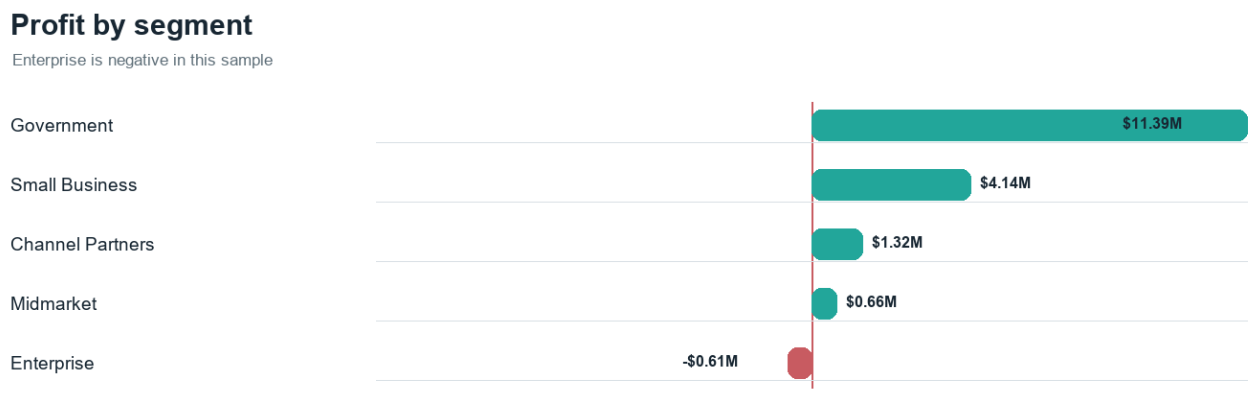


Figure 3. Profit by segment. Enterprise is the only loss-making segment in this dataset.

7. FINDINGS, INTERPRETATION AND PRESENTATION

Key observations from the supplied sample and a reproducible demonstration path



Figure 4. Profit by country. Country totals are aggregated from the source transaction rows.

Main findings

Overall: Gross Sales $127.93M less Discounts $9.21M gives Sales $118.73M; after COGS of $101.83M, Profit is $16.89M (14.2% of sales).
Product: Paseo leads profit at $4.80M.
Segment: Government has the highest profit ($11.39M); Enterprise records a loss of $0.61M and needs closer review by product, discount and country.
Country: France has the highest profit at $3.78M.
Time comparison: 2013 contains only September–December, while 2014 contains all 12 months. Do not interpret the raw annual totals as a like-for-like year-over-year comparison.

How to present your workings

1. Open the PBIX and begin on Executive Overview. **2.** Show all years, then select a year, country, segment or product to demonstrate cross-filtering. **3.** Open Profit & Loss Report and trace Gross Sales - Discounts - Sales - COGS - Profit. **4.** Open Profitability Analysis and investigate why Enterprise is negative; compare product/country results and inspect detail rows. **5.** Explain the Calendar relationship, measures and validation checks.

Limitations and source

This Microsoft demonstration dataset is not audited company data and should not drive real financial decisions. Repeated dimension/date combinations exist and there is no unique transaction ID. Use the CSV for import into Power BI; the XLSX is formatted for review. Source: Microsoft Learn - Financial Sample workbook.